1 2

Abstract

Accurate prediction of deforestation risk at pixel scale is a prerequisite for early-warning systems and targeted conservation interventions, yet the relative importance of different information sources remains poorly characterised, particularly in the data-scarce Congo Basin. We present a systematic feature ablation study across three dimensions— information type, spatial scale of neighbourhood context, and temporal depth of lag encoding—applied to a 250 000-pixel sample of Hansen Global Forest Change data over Central Africa (tile 10°N/020°E, 2016–2024). An XGBoost gradient boosted trees classifier is trained on a strict forward-looking temporal split (training predicts 2020–2022, validation predicts 2024) to avoid any form of data leakage. The core finding is that *spatial contagion*—the deforestation rate measured within 150 m to 5 km of each pixel—is the dominant predictive signal, outperforming demographic, climatic, and economic variables by a substantial margin (AUC-ROC 0.947 vs. $\leq$0.86 for all other single-group models). Temporal depth beyond a single lag year provides marginal gain (ΔAUC $<$0.003), revealing short-term stationarity in deforestation dynamics. A four-radius buffer combination achieves AUC-ROC 0.957 and PR-AUC 0.076 (baseline prevalence: 0.36%), yielding a 23-fold lift over random screening at the 1% threshold. We discuss implications for operational monitoring and the limits of governance and economic indicators as predictive features.

Keywords: deforestation, Congo Basin, machine learning, XGBoost, spatial contagion, feature ablation, early warning

1 Introduction

The Congo Basin harbours the world's second-largest contiguous tropical forest and is critical for global carbon storage and biodiversity. Yet it is undergoing accelerating tree cover loss, with annual deforestation rates estimated at 0.2–0.4% driven predominantly by small-scale subsistence agriculture [1, 2]. Unlike South-East Asian and Brazilian deforestation, which are concentrated in large industrial clearings, Congo Basin forest loss is spatially diffuse [3, 4], making remote-sensing-based detection challenging but early-warning prediction particularly valuable.

Pixel-scale prediction of where deforestation will occur one to two years ahead would allow NGOs, national forest agencies, and REDD+ programmes to target patrolling and land-tenure interventions more efficiently. The literature on machine learning for deforestation prediction has grown substantially, encompassing random forests on tabular features, convolutional neural networks on image patches [5, 6], and, more recently, large

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pre-trained Earth-observation models [7, 8]. Yet few studies have systematically decomposed predictive performance to answer the practical question: *which information sources and encoding choices actually matter?*

Drivers of deforestation span multiple spatial and temporal scales. Distance to roads and settlements determines access cost; land-tenure and governance shape incentives [9, 10]; commodity prices influence opportunity cost at national scale [11, 12]; and local deforestation history encodes a spatial contagion process by which cleared land attracts further clearing at the forest frontier [13]. It remains unclear how these factors trade off in predictive power, or whether the predictive signal is sufficiently stationary to be captured with a short temporal window.

Here we address these gaps through a triple ablation experiment on a 250 000-pixel dataset spanning Central Africa (2016–2024). We evaluate nine categories of predictor variables extracted from thirteen open data sources via Google Earth Engine [14]; test four spatial scales for neighbourhood deforestation context (150 m, 500 m, 1.5 km, 5 km radii); and vary temporal encoding from one to four lag years. All experiments use a strict forward-looking temporal split to ensure that training data never overlaps with the validation forecast window, a design criterion often violated in prior work [15].

2 Data and methods

2.1 Study area and target variable

The study area corresponds to the Hansen Global Forest Change (GFC v1.12) [16] tile at 10°N/020°E, covering approximately 10° × 10° across the Democratic Republic of Congo, Central African Republic, South Sudan, Chad, and Uganda (Figure 1). We sampled 250 000 pixels at 30 m resolution using stratified random sampling, oversampling forest pixels with recent loss history. The binary target is $y = 1$ if pixel i experienced tree-cover loss in year t (the *prediction year*), and $y = 0$ otherwise, as recorded in the Hansen `lossyear` band.

2.2 Feature engineering and data sources

For each pixel i and each prediction year t^*, we extract predictor features from the preceding K years using a *sliding window* encoding. Let $x_{i,t}$ denote the value of a time-varying

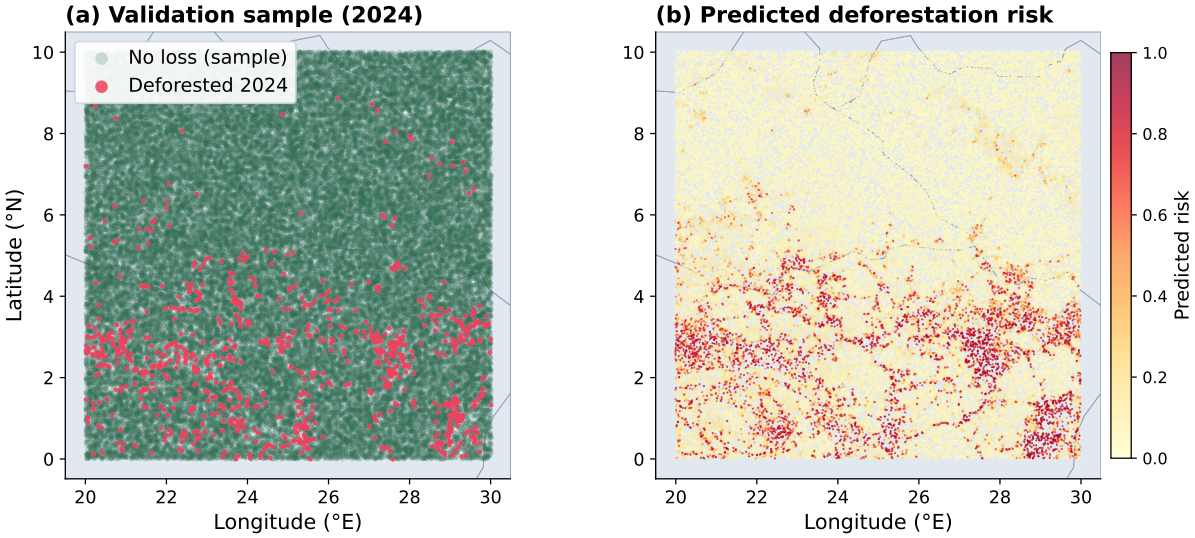

Figure 1: (a) Spatial distribution of the 240 230 validation sample pixels (2024 prediction year) over the 10°N/020°E Hansen tile. Red dots indicate pixels that experienced tree-cover loss in 2024 (871 events, 0.36% prevalence). (b) Predicted deforestation risk for the same pixels from the full 549-feature model (AUC-ROC = 0.954); this model is used for the spatial risk map because it was deployed in the operational monitoring application. The parsimonious 139-feature core model achieves AUC = 0.955 with higher PR-AUC (see text). Elevated risk (orange–red) clusters along the eastern DRC forest frontier and the Central African Republic–DRC border. Country boundaries from Natural Earth.

variable for pixel i in year t; three complementary representations are derived:

$$\text{(i) lags:} \quad x_{i,t^*-k}, \quad k = 1, \ldots, K, \tag{1}$$

$$\text{(ii) differences:} \quad \Delta x_{i,t^*-k} = x_{i,t^*-k} - x_{i,t^*-k-1}, \quad k = 1, \ldots, K-1, \tag{2}$$

$$\text{(iii) window mean:} \quad \bar{x}_{i,t^*} = \frac{1}{K} \sum_{k=1}^{K} x_{i,t^*-k}. \tag{3}$$

Because all representations use only values strictly before t^*, no future information leaks into the features. This encoding is applied to all time-varying sources; the ablation over encoding schemes (Section 3.3) evaluates each representation individually and in combination.

Static or slowly-varying features are extracted once per pixel: topography from the Shuttle Radar Topography Mission [17]; distance to the nearest road, settlement, and protected area from OpenStreetMap [18]; and binary protected-area status from the WDPA database enriched with IUCN governance categories.

Time-varying sources include: multi-scale neighbourhood deforestation rates at 150 m, 500 m, 1.5 km and 5 km radii computed from the Hansen GFC raster (the *spatial contagion* features); pixel-level cumulative forest loss; annual population density from WorldPop [19]; monthly precipitation from CHIRPS [20]; temperature, evapotranspiration, and

soil moisture from ERA5-Land [21]; night-time light radiance from VIIRS [22]; MODIS active fire counts [23]; and three global commodity prices (cocoa, palm oil, gold) as proxies for economic pressure. After sliding-window expansion, the full feature set comprises up to 549 variables organised into the nine groups listed in Table 1.

2.3 Experimental design

Temporal split. To evaluate true out-of-sample generalisation we construct two physically separate datasets. The *training set* covers prediction years 2020–2022 (with features computed from 2016 onward) and contains $N_{\text{train}} = 729\,114$ pixels, of which $n^{+}_{\text{train}} = 2\,525$ experienced deforestation (prevalence 0.35%), where n^{+} denotes the number of positive (deforested) samples. The *validation set* targets prediction year 2024 (features from 2017 onward; $N_{\text{val}} = 240\,230$, $n^{+}_{\text{val}} = 871$, prevalence 0.36%). By construction, the training set never includes any observation from 2023, which is reserved exclusively as lag-1 input for the 2024 validation predictions.

Classifier. We train an XGBoost gradient-boosted tree ensemble [24] with early stopping on validation log-loss. To handle the severe class imbalance, we set the `scale_pos_weight` parameter to the ratio of negative to positive training samples (≈288), which up-weights the minority class proportionally. Hyperparameters are tuned via Bayesian optimisation (50 trials, Optuna [25]) minimising validation log-loss; the best configuration is then retrained on the full training set. All generalisation claims rest on the held-out 2024 forecast; no spatial or temporal cross-validation is used during training.

Ablation experiments. We conduct three sets of ablation experiments. *Dimensional ablation*: each of the nine feature groups is used alone, and groups are then added cumulatively in decreasing order of single-group AUC. *Spatial ablation*: the four buffer radii are evaluated in selected pairwise combinations and as a full set. *Temporal ablation*: window depth is varied from $K = 1$ to $K = 4$ years, and four encoding schemes are compared (lags only; lags + summaries; lags + differences; full encoding).

Evaluation metrics. We use two complementary ranking metrics. The *area under the receiver operating characteristic curve* (AUC-ROC) measures a classifier's ability to rank positive examples above negative ones across all decision thresholds: a random classifier scores 0.5, a perfect classifier scores 1.0. Because AUC-ROC can appear misleadingly high when positives are rare [26], we also report the *area under the precision–recall curve* (PR-AUC), which focuses on the model's ability to maintain high precision as recall increases. For PR-AUC the baseline of a random classifier equals the class prevalence—here approximately 0.004—so any value substantially above this threshold indicates genuine predictive skill.

We complement these with two operationally interpretable quantities. The *concentration curve* $C(s)$ gives the fraction of total deforestation events recovered when screening the top-s fraction of pixels ranked by predicted risk; it measures how tightly the model concentrates risk into a small target area. The *lift* at screening fraction s is defined as $\mathcal{L}(s) = \text{recall}(s)/s$; a lift of 23 at $s = 0.01$ means the top 1% of predicted pixels contains 23 times more deforestation events than a random selection of 1% of pixels would.

Interpretability. TreeSHAP [27, 28] is used to attribute model predictions to individual features, enabling identification of the dominant predictive mechanisms.

3 Results

3.1 Dimensional ablation: spatial contagion dominates

Table 1 summarises the performance of each feature group in isolation. The spatial contagion group achieves by far the highest single-group performance (AUC-ROC 0.947, PR-AUC 0.051 with 100 features), followed at a substantial distance by infrastructure access (AUC 0.858, 12 features) and population density (AUC 0.832, 13 features). Local Hansen deforestation history without neighbourhood context yields AUC 0.792, confirming that the landscape-scale signal—where is the deforestation frontier?—is more informative than the pixel's own past. Economy (commodity prices, governance indices, WDI) performs no better than chance (AUC 0.500), a result we discuss in Section 4.

Cumulative combinations show rapid saturation. Combining spatial contagion features with Hansen local history already reaches AUC 0.955, PR-AUC 0.075 (127 features). Adding infrastructure access produces the optimal configuration (AUC 0.955, PR-AUC **0.076**, 139 features; hereafter the *core model*). Beyond this core, adding population, climate, NTL, and economic features leaves AUC unchanged while PR-AUC decreases from 0.076 to 0.068. This pattern is consistent with noise dilution in the minority class: adding features that are weakly correlated with the target increases the variance of tree splits without improving signal, causing the boosted ensemble to fit spurious patterns in the positive class and thereby degrading precision at high recall thresholds. AUC-ROC is less sensitive to this effect because it evaluates ranking globally across both classes, whereas PR-AUC weights errors in the rare positive class more heavily.

3.2 Spatial ablation: multi-scale buffers improve minority recall

Figure 2(b) and Table 2 compare configurations of the four buffer radii. In single-radius configurations, the 500 m buffer yields the highest AUC-ROC (0.944), while the 150 m buffer provides the highest PR-AUC (0.072), suggesting that micro-scale contagion (clearing events within a single Landsat pixel-neighbourhood) carries disproportionate signal

Table 1: Dimensional ablation: validation performance by feature group (sorted by single-group AUC-ROC). Cumulative results show the core model (Hansen + Spatial + Infra) as the optimal configuration.

Feature group	N_{feat}	AUC-ROC	PR-AUC
Single-group models			
Spatial contagion (buffers)	100	0.947	0.051
Infrastructure & protection	12	0.858	0.024
Population density	13	0.832	0.017
Hansen local history	27	0.792	0.013
Climate (ERA5-Land)	176	0.773	0.010
Country	7	0.737	0.007
Precipitation (CHIRPS)	39	0.712	0.007
Night-time lights (VIIRS)	100	0.666	0.006
Economy (prices, governance)	75	0.500	0.004
Cumulative combinations			
Hansen + Spatial	127	0.955	0.075
Core: + Infrastructure	**139**	**0.955**	**0.076**
Core + Population	152	0.956	0.073
All features	549	0.954	0.068

for the rarest, most localised events. The 5 km radius performs comparably in AUC but substantially below in PR-AUC (0.042), consistent with the interpretation that regional deforestation rates are correlated predictors but lack the specificity needed to identify individual at-risk pixels.

Combining radii improves upon any single-radius model. The full four-radius combination achieves the highest AUC (0.957), and pairing the 150 m with the 1.5 km radius reaches AUC 0.948 and PR-AUC 0.072 with only 89 features, suggesting that coarse- and fine-scale contagion carry complementary information.

3.3 Temporal ablation: one lag year is sufficient

Table 3 shows that extending the temporal window from one to four lag years yields negligible improvement in both metrics (ΔAUC < 0.003, ΔPR-AUC < 0.002), with the single-year model already achieving AUC-ROC 0.954 and PR-AUC 0.076. This result is consistent across encoding schemes: the lags-only representation achieves AUC 0.953 while the full encoding (lags + differences + summaries) reaches AUC 0.956, a difference of 0.003 that is unlikely to be operationally meaningful. The finding implies that the deforestation process in this region is effectively first-order Markov: the current frontier location, captured by the most recent neighbourhood deforestation rate, encodes nearly all the predictive information available in the time series. Adding window summaries

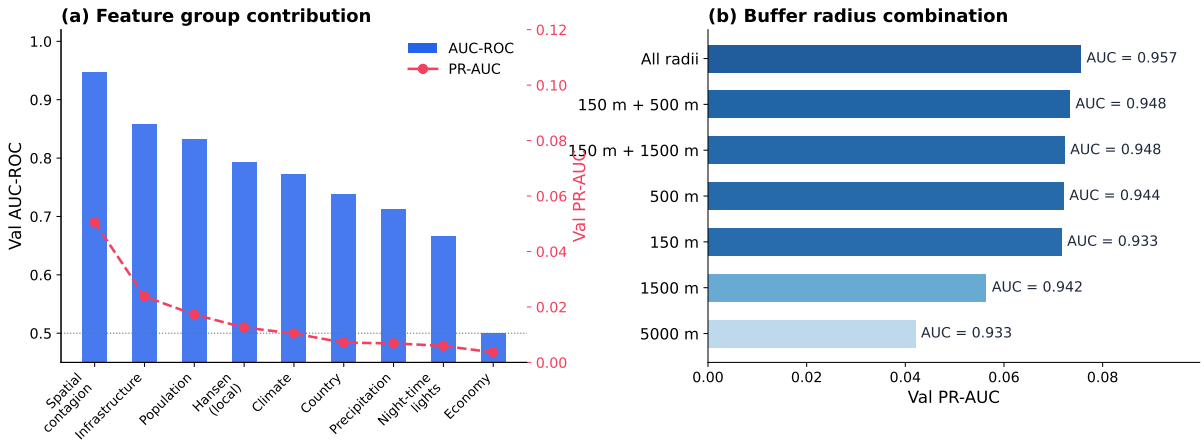

Figure 2: Feature ablation results on the 2024 validation set. (a) Dimensional ablation: validation AUC-ROC (blue bars, left axis) and PR-AUC (orange dots, right axis) for each feature group used alone, ordered by single-group contribution. (b) Spatial ablation: PR-AUC for seven buffer-radius configurations; AUC-ROC annotated on each bar.

Table 2: Spatial ablation: validation performance for selected buffer radius configurations. All models use the full 4-year temporal window and Hansen + infrastructure contextual features.

Configuration	N_{feat}	AUC-ROC	PR-AUC
150 m only	64	0.933	0.072
500 m only	64	0.944	0.072
1500 m only	64	0.942	0.056
5000 m only	64	0.933	0.042
150 m + 500 m	89	0.948	0.073
150 m + 1500 m	89	0.948	0.072
150 m + 5000 m	89	0.949	0.073
All radii	**139**	**0.957**	**0.076**

to raw lags provides a modest gain over lags alone (AUC 0.956 vs. 0.953), while further adding delta-encoded trend terms shows no consistent improvement.

3.4 Screening efficiency and interpretability

Figure 3(a) shows the concentration curve for the deployed model on the 2024 validation set. Prioritising the 1% of pixels with the highest predicted risk captures 22.7% of all deforestation events—a 23-fold improvement over random area selection. Screening 10% of pixels recovers 90.9% of the observed deforestation, confirming that the model concentrates the signal effectively. This translates to a nine-fold improvement in patrol efficiency over random area allocation.

TreeSHAP attribution (Figure 3(b)) identifies the window-mean deforestation rate at

Table 3: Temporal ablation: validation metrics as a function of lag-window depth (core feature set).

Window depth	Features	AUC-ROC	PR-AUC
1 year	29	0.954	0.076
2 years	104	0.957	0.075
3 years	119	0.955	0.077
4 years	139	0.957	0.076

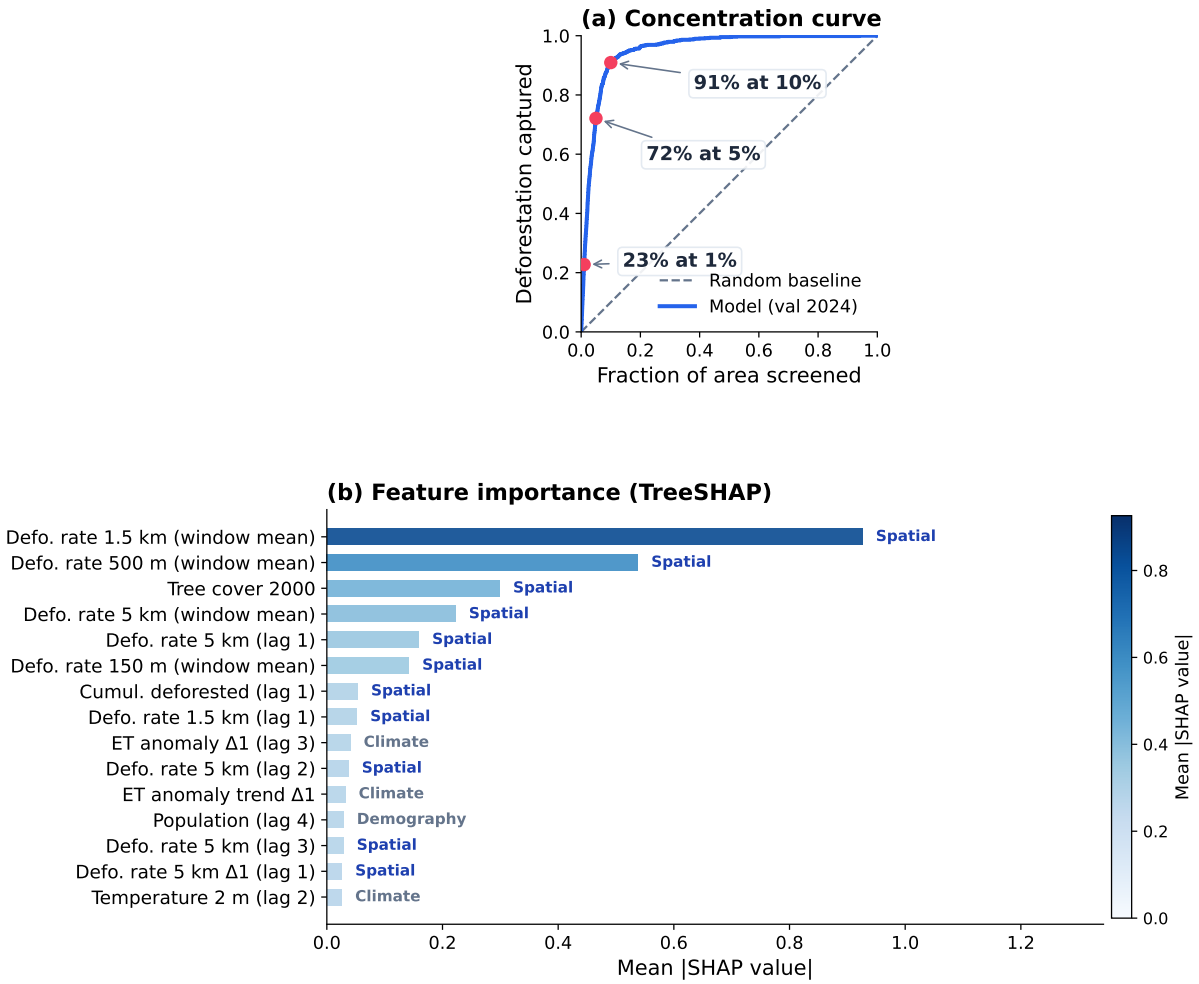

Figure 3: (a) Concentration curve for the deployed model on the 2024 validation set: fraction of observed deforestation captured versus fraction of area screened. Annotated values indicate recall at 1%, 5%, and 10% screening thresholds. The dashed line represents random screening. (b) Top-15 features by mean absolute SHAP value. Bar colour encodes magnitude (sequential blue gradient: darker = higher importance). Feature group labels (Spatial, Climate, Demography) are annotated to the right of each bar.

1.5 km radius (`defo_rate_1500m_wmean`) as the dominant feature by a wide margin, followed by the 500 m rate and the pixel's 2000 tree-cover baseline. The top-6 features are all spatial contagion variables or the static forest-cover baseline, confirming the dimensional ablation findings. The first non-spatial feature to appear is evapotranspiration anomaly at lag 3 (position 9), which may reflect drought-induced stress as a permissive condition for clearing activity. This lag-3 climate feature is present because the SHAP values are computed within the full four-year model and reflect marginal contributions conditional on all other features; the temporal ablation instead tests whether dropping entire lag years from a fresh model degrades performance, and finds negligible degradation. The two findings are therefore complementary: a three-year-old evapotranspiration signal carries some residual information not captured by spatial contagion alone, yet a model trained with only one lag year captures nearly all predictive performance because the spatial signal is sufficiently dominant.

All reported metrics are point estimates on the held-out 2024 validation set; bootstrap confidence intervals were not computed. Metric differences smaller than approximately 0.005 in AUC-ROC or 0.003 in PR-AUC should therefore be interpreted cautiously, as they may fall within sampling variability.

High-resolution forecast. To illustrate operational applicability, we extract a dense 250 m grid (13,232 points) within a $0.3^\circ \times 0.3^\circ$ zone in northeastern Congo and apply the trained model to predict 2025 deforestation risk (Figure 4). Since Hansen GFC v1.12 covers loss through 2024, the 2025 forecast constitutes a true out-of-sample prediction for which no ground truth yet exists. The resulting risk map reveals a spatially coherent deforestation frontier: high-risk pixels (red) concentrate along roads and existing clearings, while intact forest patches (green) remain at low predicted risk—consistent with the contagion-dominated dynamics identified by the ablation study.

4 Discussion

Why spatial contagion dominates. Deforestation in the study region is primarily driven by smallholder agriculture expanding outward from existing clearings and roads [1, 2], consistent with broader patterns documented for the Congo Basin. This process is spatially autocorrelated: the probability that a forested pixel is cleared next year depends mainly on whether its neighbours were cleared recently, not on abstract economic signals. Our results quantify this precisely: the spatial contagion group alone outperforms the combination of population, climate, and economic predictors (AUC 0.947 vs. 0.843 for the next-best group).

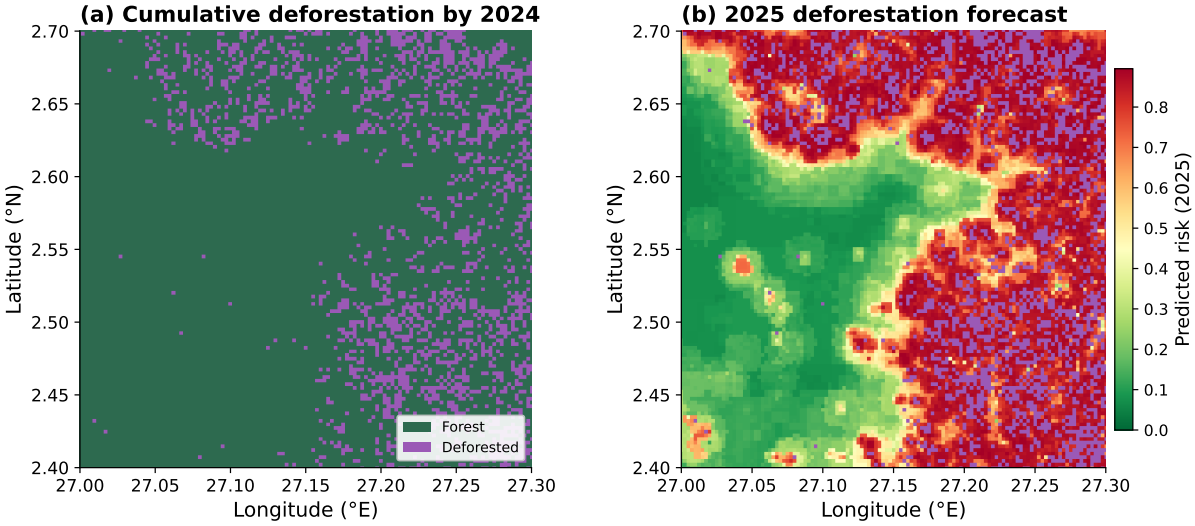

Figure 4: High-resolution deforestation map and 2025 forecast. (a) Cumulative deforestation status by 2024 at 250 m spacing (14,641 points): green = forested, violet = deforested (Hansen GFC v1.12 `lossyear` $\in [1,24]$). (b) Predicted deforestation risk for 2025 (same grid). Green-to-red gradient indicates predicted risk for currently forested pixels; violet pixels are already deforested (carried from panel a). This is a true forecast: no 2025 deforestation data exists at time of writing.

Temporal stationarity. The sufficiency of a one-year lag window is consistent with a frontier-driven process where the *current position* of the clearing front (encoded by $t-1$ neighbourhood rates) is more informative than its historical trajectory. This has practical implications: monitoring systems can be updated annually with only the most recent deforestation layer, rather than requiring a multi-year historical archive.

Governance and economic indicators. The failure of the economy group (commodity prices, WGI, WDI) to explain deforestation at pixel scale can be attributed to a fundamental resolution mismatch: WGI and WDI indices are reported at country level, while the model predicts at 30 m pixel resolution. This six-to-seven orders-of-magnitude difference in spatial scale means that governance variables enter the model as spatially constant values within each country, providing no within-country discriminative power. The model therefore receives identical governance values for millions of pixels with very different deforestation probabilities, and the classifier correctly learns to ignore a signal that is uninformative at the prediction unit. This supports the conclusion of [9] that macro-economic drivers operate at national or sub-national scale and do not translate to pixel-level risk once landscape-scale spatial contagion is accounted for. Protected-area governance type (IUCN category) provides no improvement over a binary inside/outside variable (tested separately; not shown), echoing recent critiques of governance effectiveness indicators as spatial predictors.

Limitations. No formal statistical testing was applied to the ablation comparisons. Metric differences smaller than approximately 0.005 in AUC-ROC or 0.003 in PR-AUC should be interpreted cautiously, as they may reflect sampling variability rather than genuine differences in predictive power; bootstrap confidence intervals would be needed to confirm such margins. This study is also limited to a single 10°×10° tile and a single predictand (annual binary tree-cover loss). Generalisation to other ecoregions—where deforestation may be industrially-driven rather than smallholder-driven—is untested. The 30 m spatial resolution of the GFC product may undercount small clearings, leading to underestimation of true deforestation prevalence. Because PR-AUC is sensitive to the positive class rate, this implies that the reported PR-AUC values are conservative: finer-resolution data would expose more true positives currently hidden below the 30 m detection threshold, increasing the effective positive rate and yielding higher absolute PR-AUC values even with identical discrimination ability. Future work should evaluate multi-tile generalisation and incorporate land-use transition probabilities rather than binary occurrence.

5 Conclusions

We have shown, through a systematic triple ablation study, that pixel-scale deforestation risk within the study tile (10°N/020°E, covering Central Africa) is overwhelmingly determined by spatially-encoded contagion: the local frontier position at 150 m to 5 km from each pixel. A parsimonious 139-feature core model—combining multi-scale neighbourhood deforestation history with topographic and access-cost predictors—matches the AUC of a 549-feature model while improving minority-class recall (PR-AUC 0.076 vs. 0.068), recovering 91% of observed deforestation by screening only 10% of the study area. Temporal depth beyond a single lag year provides no meaningful gain, suggesting that annual updates of a lightweight model suffice for operational early-warning applications. Inference on 250 000 pixels completes in under one second on a standard laptop without GPU, making the approach suitable for deployment in resource-constrained settings. These results provide a principled basis for feature selection in applied deforestation monitoring systems and highlight the practical limitations of governance and economic indicators as pixel-scale predictors.

Acknowledgements

The authors thank the Google Earth Engine team for providing computational access to the Hansen Global Forest Change dataset and ancillary geospatial layers. Hansen GFC data are freely available from the Global Land Analysis and Discovery laboratory (GLAD, University of Maryland). Population data were obtained from WorldPop (University of

Southampton). Road and settlement data are from OpenStreetMap contributors under the Open Database Licence. Protected-area boundaries are from the World Database on Protected Areas (WDPA, UNEP-WCMC / IUCN). No dedicated funding was received for this work.

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